Data science for business project

Introduction

The aim of this project is to estimate the state of the electricity grid in Finland in order to forecast when during a given week would be a good time to sell or buy electricity based on the estimation of the grid state. When the grid is in surplus, meaning it is producing more electricity than is consumed, it would be a good time to buy electricity since the price is lower. On the other hand, if the grid is low on power it would be a good time to sell into the grid. Such prediction would be beneficial to small businesses and private individuals who own solar panels or power storages, such as backup battery storages. The estimation of this "net imbalance volume" or grid state is estimated using datasets from Fingrid containing data of solar, wind power forecasts, real time nuclear power outputs as well as temperatures in Helsinki and imbalance price. The imbalance price is the balancing price for the energy determined for every quarterly hour. When the grid is in balance the price for the energy is the estimated imbalance price. During the times of negative imbalance, or shortage of energy, the imbalance price goes up, because the grid needs more power and producers who can sell gets paid "premium" to balance the energy system. On the other hand, when the grid is in surplus, or too much energy is produced than consumed, there is a need to sell at a lower than market price. Thus, the imbalance price is lower than estimated.
This gives the motivation for this project, to train a machine learning model with the datasets loaded from Fingrid that would give accurate estimations on when would be profitable to buy energy cheap and sell when price is higher. After training, the model is then given a week of unseen data, where it tries to predict the abovementioned times by hour, and then compared to factual data of the realized imbalance power and calculated would it have been profitable to follow the suggestions of the machine learning model.

Hypothesis

The goal in this project is to train a ML model to estimate if the imbalance price deviates enough from "normal price" to be profitable to buy cheap and sell high, given the model a week long data of wind and solar forecast, nuclear power outputs and temperature. The hypothesis is that there should be positive correlation between the energy outputs and the temperature that impacts the imbalance price, similar to the effects seen on the energy spot price spiking. Such conditions would be low solar and wind production combined with a very cold, lower than -20 degrees Celsius, temperature that would drive the imbalance price up as there is more need for energy. Furthermore, should abovementioned correlations exists, they would be usable in predicting profitable times to buy and sell energy.

Limitations

Energy price marker is very volatile, which makes predicting the energy price difficult. Estimating the imbalance price consists of multiple variables, some of which have limited access, that further complicates making accurate estimation. In this project,

only few variables are taken into consideration and the dataset used for the machine learning model spans only 2 years from 2024 to 2026. Therefore, this experiment is poses a question, if with this limited amount of data it is possible to make credible estimation if the imbalance price is high or low enough to buy or sell.

Datasets and data preprocessing

Initial datasets for training the models contains 183135 and 725219 rows of 5 different features, which require quite a lot of preprocessing, since these two datasets has data in two varying intervals. Longer dataset contains data in 15 minute intervals and the shorter one in 1 hour interval. Effectively, both datasets are combined and filtered into 1 hour interval in order for the data to be consistent throughout the dataset. The combined training dataset timeline spans from 26.2.2024 to 27.2.2026.

```
datasetId         Wind_forecast  Solar_forecast  Price  Temperature  \
DateTime
2024-02-26 22:00       1013.250           0.000  58.88     0.766530
2024-02-26 23:00        791.375           0.000  51.00     0.766530
2024-02-27 00:00        698.375           0.000  50.00     0.766530
2024-02-27 01:00        754.350           0.000  48.00     0.451566
2024-02-27 02:00        812.325           0.000  46.00     0.227575
2024-02-27 03:00        831.000           0.000  46.00     0.185530
2024-02-27 04:00        851.525           0.125  55.68     0.185530
2024-02-27 05:00        834.000           0.225  91.37     0.108316
2024-02-27 06:00        786.500          16.375  96.23    -0.329230
2024-02-27 07:00        694.200          58.400  51.50     0.101169

datasetId         Nuclear_power
DateTime
2024-02-26 22:00    4366.424998
2024-02-26 23:00    4363.784999
2024-02-27 00:00    4364.199997
2024-02-27 01:00    4364.689999
2024-02-27 02:00    4363.679998
2024-02-27 03:00    4364.729999
2024-02-27 04:00    4363.084998
2024-02-27 05:00    4363.129998
2024-02-27 06:00    4363.079998
2024-02-27 07:00    4363.524998
Number of NaN in Temperature:  40
Number of NaN in wind:  0
Number of NaN in solar:  0
Number of NaN in nuclear:  40
Number of NaN in price:  0

table size:  17568 rows,  87840 datapoints
Showing 40 rows with missing data:
```

Figure 1

After combining the dataset together, there's the required step to handle the NaN value rows. In this dataset, out of 17568 rows only 40 contain NaN values, which is not a significant number even if all of them are deleted from the dataset. However, that is not the ideal case and after examining the NaN rows only 29 of them contain are in sequential rows of more than 2 consecutive rows. For the 11 individual rows the values could be interpolated based on the surrounding values.

```
datasetId         Wind_forecast  Solar_forecast  Price  Temperature  \
DateTime
2024-02-26 22:00       1013.250             0.0  58.88     0.766530
2024-02-26 23:00        791.375             0.0  51.00     0.766530
2024-02-27 00:00        698.375             0.0  50.00     0.766530
2024-02-27 01:00        754.350             0.0  48.00     0.451566
2024-02-27 02:00        812.325             0.0  46.00     0.227575

datasetId         Nuclear_power
DateTime
2024-02-26 22:00    4366.424998
2024-02-26 23:00    4363.784999
2024-02-27 00:00    4364.199997
2024-02-27 01:00    4364.689999
2024-02-27 02:00    4363.679998
datasetId
Wind_forecast      0
Solar_forecast     0
Price              0
Temperature       29
Nuclear_power     29
dtype: int64
Showing 29 rows with missing data:
```

Figure 2

Thus, only 29 rows are left with NaN-values. This is only ~0.17% of the data rows, so minimal loss of data after deleting them.

Visualizations

After initial preprocessing, it's time to visualize the data. For each variable there's line chart to examine the seasonal variations throughout the timeline (figure 3). To simplify the visualization the values are averaged to 24h mean values and shown along with a one week moving average.

Figure 3

The second visualization (figure 4) examines daily rhythm of each variable. This boxplot highlights some of the predictable phenomena in the data, such as the solar and temperature rhythms, that are naturally higher during the daytime. Thus, it aligns with the seasonal variations from the previous figure (figure 3). Furthermore, figure 4 points out the steady nature of the nuclear power production, which could be seen as quite erratic and unpredictable in figure 3. Wind production is another very volatile variable and that reflects to the overall mean being on the lower side throughout the timeline with some exceptional peaks on windier occasions. Some seasonal fluctuations (figure 3) is noticeable in that also in wind forecast, where higher variance is seen on the autumn-winter months compared to summer months. Lastly, examining the price graphs, it appears to be rather independent from the other variables, even though could assume that the other variables would, at least to some degree, explain the peaks and higher variance in the imbalance price.
Nevertheless, based on these variables and the other features engineered next, the models will try to catch the underlying patterns and then test would it be profitable to buy and sell during the suggested times.

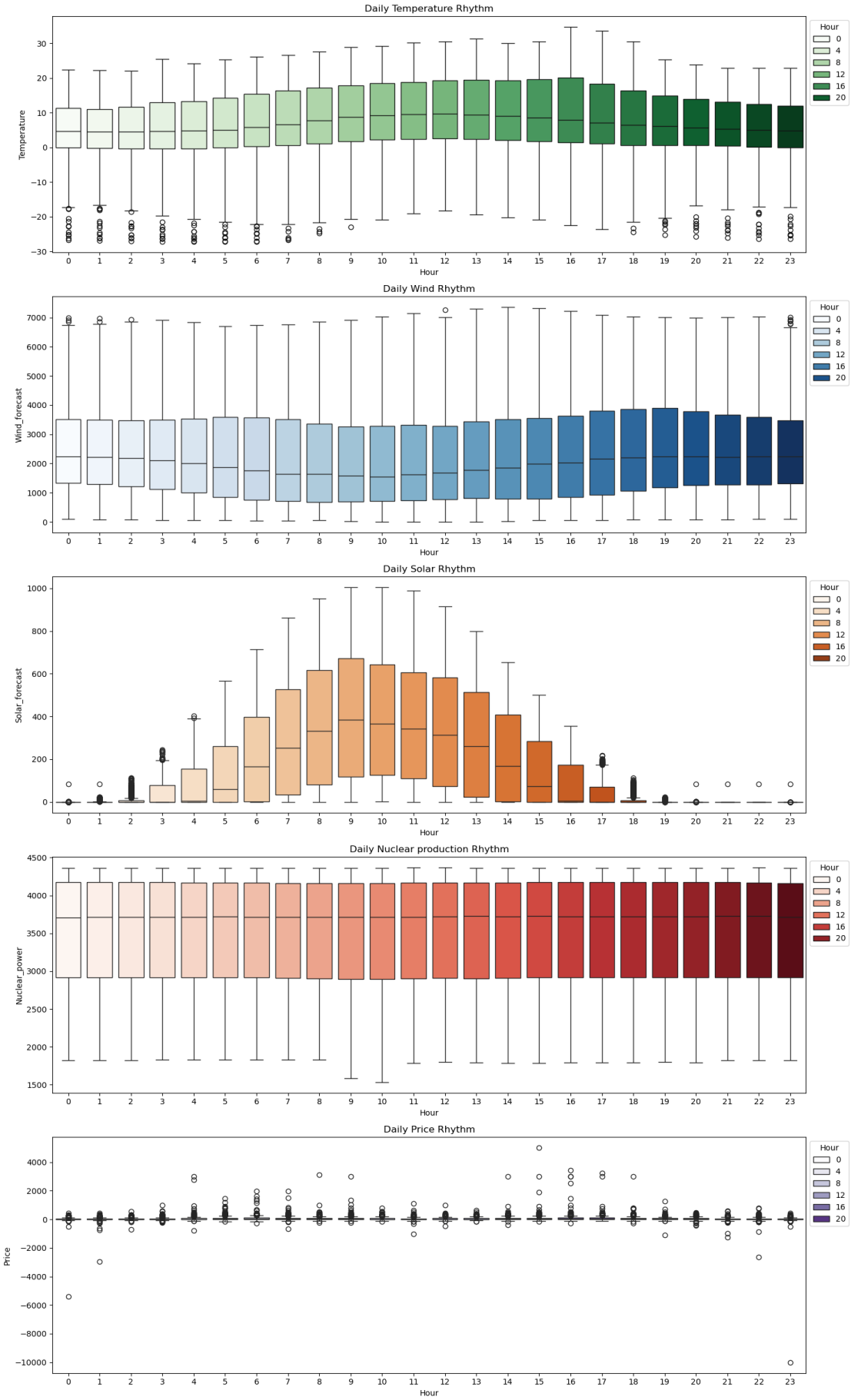
Daily Temperature Rhythm
Temperature
Hour
Daily Wind Rhythm
Wind_forecast
Hour
Daily Solar Rhythm
Solar_forecast
Hour
Daily Nuclear production Rhythm
Nuclear_power
Hour
Daily Price Rhythm
Price
Hour
Hour
0
4
8
12
16
20

Figure 4

Feature engineering

Next steps are to prepare the dataset for the machine learning models. This is a classification problem, where we have three possible outcomes from the model. In trading terms, they are called “long”, “balanced” and “short”, and will be determined so that when the price deviates from the imbalance price by at least 30% or 20€, whichever is smaller, it is time to buy or sell. For the model, these labels are set as simple numbers from 0 to 2 respectively. Furthermore, the dataset requires braking down the datetime into individual columns for day of the week, hour, month and year, since the machine learning models doesn’t understand the datetime value as calendar time. Lastly, the datasets require setting timeframe lags, since the dataset is a time series. One hour lags were created for the price and each power outputs, 4 columns in total, and 24h lags for temperature and price. These allow the model to better understand how the current values are linked to the previous ones and have higher chance in delivering good performance.
After preprocessing and feature engineering, the dataset (figure 5) that goes for training the data contains 17 features with 17323 rows of observations.
These same steps are also executed for the forecasting dataset, which was loaded separately and is not part of the training set.

DateTime	Wind_forecast	Solar_forecast	Price	Temperature	Nuclear_power	Day_of_week	Hour	Month	
2024-03-08 22:00:00	192.1	0.0	50.0	-2.8674	2796.099995	4	22	3	
2024-03-08 23:00:00	210.175	0.0	45.34	-3.037263333333333	2796.23333	4	23	3	
2024-03-09 00:00:00	228.3	0.0	44.0	-3.207126666666667	2796.366665	5	0	3	
2024-03-09 01:00:00	260.425	0.0	38.22999	-3.37699	2796.5	5	1	3	
2024-03-09 02:00:00	312.175	0.0	38.22999	-3.4269350000000003	2796.9699950000004	5	2	3	
	Year	Rolling median	Target	Price_diff_1h	Price_diff_24h	Wind_forecast_diff_1h	Nuclear_power_diff_1h	Solar_forecast_diff_1h	Temperature_lag24
2024-03-08 22:00:00	2024	64.905	1.0	3.0	-68.88	-36.0	0.22499999999990905	0.0	5.526296
2024-03-08 23:00:00	2024	64.905	1.0	-4.659999999999997	-71.66	18.075000000000017	0.1333349999999882	0.0	4.337106166666667
2024-03-09 00:00:00	2024	64.905	0.0	-1.3400000000000034	-20.89	18.125	0.1333349999999882	0.0	3.0195478333333328
2024-03-09 01:00:00	2024	64.905	0.0	-5.770009999999999	-26.640010000000004	32.125	0.1333349999999882	0.0	2.251201
2024-03-09 02:00:00	2024	64.905	0.0	0.0	-78.77001	51.75	0.46999500000038097	0.0	1.7127315000000003

Figure 5

Before training of the models, one last step was ensuring the feature lags provide unique predictive values in order to avoid overfitting the model. Should the correlation matrix(figure 6) show the reds outside the diagonal then the lags would need to be adjusted.

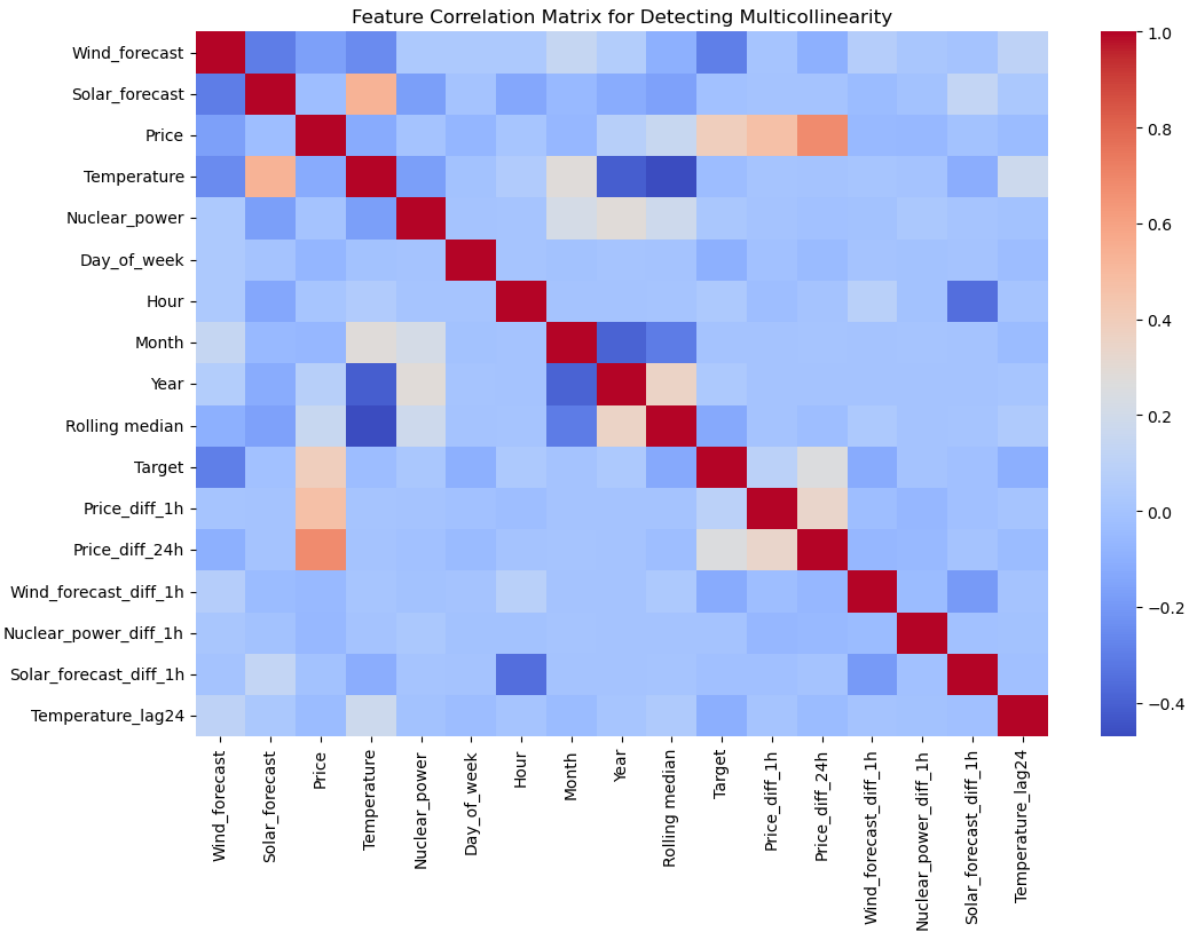

Figure 6

Training and selecting model

For this multi-label classification task with a dataset of over 80000 data points, a few models were manually selected for comparison. Logistic regression is natural number one for the classification tasks as well as the support vector classifier. Combine these two with Extreme-Gradient-Boost since the electricity market is very volatile, the boost algorithm with multiple models combined would be reasonable to try. Lastly, since the dataset is also big enough, a neural network based multi-layer perceptron was the fourth choice. The training and selection was automated with grid search, where each model had pipelines with given scalers and hyperparameters to loop through. The winning model was chosen based on the F1 score (Figure 7).

	Model	F1-Score (Macro)	Accuracy	Precision	Fit Time (s)
1	XGBoost Classifier	0.615679	0.624490	0.627819	0.801261
2	Multi-Layer-Percetron Classifier	0.501259	0.538153	0.545213	7.319140
3	Support Vector Classifier	0.486553	0.516154	0.522136	6.438371
0	Logistic Regression	0.460762	0.502525	0.498132	0.078093

Figure 7

Prediction

For the best performing model, the XGBoost classifier, a new dataset consisting of data for 28.2.2026 – 7.3.2026 was used for predicting the high and low prices for this time. Figure 8 shows in green and red these times, that would be then used to sell and buy on the respective times.

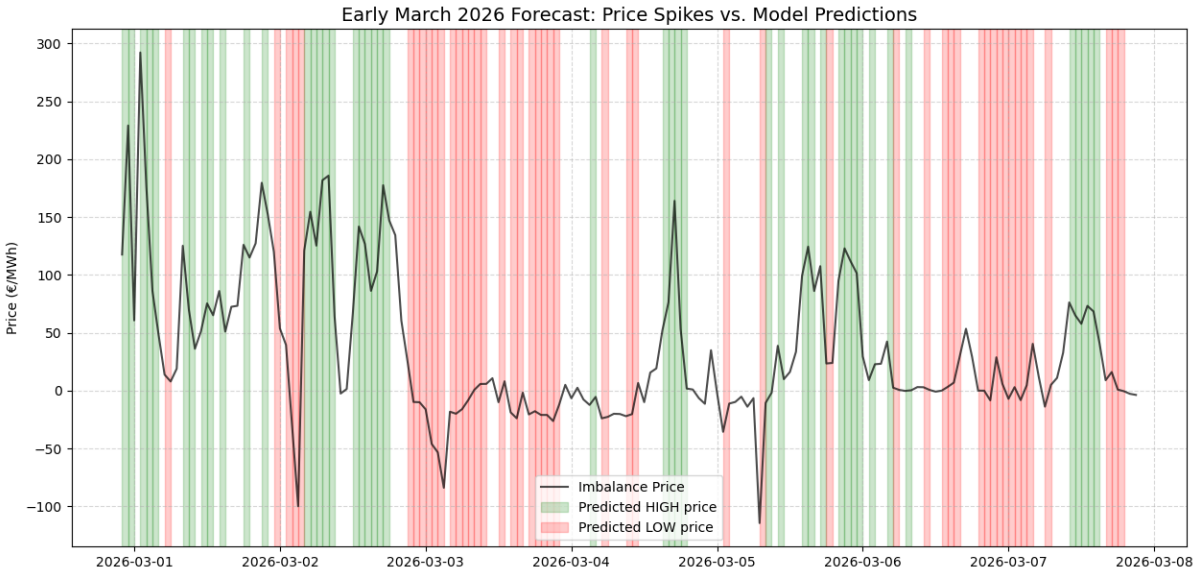

Figure 8

The predictions for high price, times to sell, were 26,79% of the labels, and low prices or time to buy were 29,17%. Together the predictions would be 55,95% during the week period or 94 trades out of the 168 hours in the week (figure 9 & 11). Lastly, evaluation on how much profit could be averagely profited, if electricity is sold to grid every time the prediction shows high and bought when prediction is low. Calculations are made with the average price of the during buy(low) and sell(high) labels (figure 10). With the total of 94 trades the profit would be for one week about 5030,42€. However, this would require for the trade size to be 1 megawatt per trade.

```
26.79 %
29.17 %
Trade predictions: 55.95%
Balanced, non-trade predictions  44.05%
```

Figure 9

```
--- Economic Validation of Labels ---
Avg Price during 'Buy' (-1):  -8.76 €
Avg Price during 'Balanced' (0): 22.12 €
Avg Price during 'Sell' (+1):  102.25 €
```

Figure 10

```
Total buy(-1) trades: 49
Total sell(1) trades: 45
Total Potential trades in March Week: 94
Potential Net Profit from Strategy, for 1MW per trade: 5030.42 €
```

Figure 11

Conclusions

For such a highly volatile market as the electricity market in Finland, the classifier shows potential for being accurate enough to predict the times of deviation from the "normal" imbalance price. The trade volumes, however, might require to be too high for it to be profitable for individuals or even small businesses.
The positive correlations between the imbalance price, the production outputs and temperature conditions were not obvious in the visualizations, though they could be further inspected on shorter timelines. Therefore, just basing buy or sell signals with the extreme weather conditions is not a feasible strategy to buy or sell electricity.

Recommendations

For the model training more hyperparameters could have been explored even more thoroughly, and gridsearch could have also be replaced with random search. Furthermore, the models could have been compared further by building the winnings estimations for each model or for the winning model with different parameters to get more comparing data for evaluation.

Codebase

https://github.com/PRoutamaa/DS_Business_project